

AN INTRODUCTION TO PHARMACOGNOSY

Introduction:

The word "pharmacognosy" is derived from two Greek words: *pharmakon* (drug), and *gnosis* (knowledge) or the Latin verb *cognosco* (*con*, 'with', and *gnōscō*, 'know'; itself a cognate of the Greek meaning 'I know, perceive'), meaning 'to conceptualize' or 'to recognize'.

The term "pharmacognosy" was used for the first time by the Austrian physician Schmidt in 1811 and 1815 by Crr. Anotheus Seydler in work titled *Analecta Pharmacognostica*.

Originally—during the 19th century and the beginning of the 20th century—"pharmacognosy" was used to define the branch of medicine or commodity sciences (*Warenkunde* in German) which deals with drugs in their crude, or unprepared, form. Crude drugs are the dried, unprepared material of plant, animal or mineral origin, used for medicine. The study of these materials under the name *pharmakognosie* was first developed in German-speaking areas of Europe, while other language areas often used the older term *materia medica* taken from the works of Galen and Dioscorides. In German the term *drogenkunde* ("science of crude drugs") is also used synonymously.

As late as the beginning of the 20th century, the subject had developed mainly on the botanical side, being particularly concerned with the description and identification of drugs both in their whole state and in powder form. Such branches of pharmacognosy are still of fundamental importance, particularly for pharmacopoeial identification and quality control purposes, but rapid development in other areas has enormously expanded the subject. The advent of the 21st century brought a renaissance of pharmacognosy and its conventional botanical approach has been broadened up to molecular and metabolomic level.

In addition to the previously mentioned definition, the American Society of Pharmacognosy also defines pharmacognosy as "the study of natural product molecules (typically secondary metabolites) that are useful for their medicinal, ecological, gustatory, or other functional properties." Other definitions are more encompassing, drawing on a broad spectrum of biological subjects, including botany, ethnobotany, marine biology, microbiology, herbal medicine, chemistry, biotechnology, phytochemistry, pharmacology, pharmaceutics, clinical pharmacy and pharmacy practice.

- **Medical Ethnobotany:** the study of the traditional use of plants for medicinal purposes;
- **Ethnopharmacology:** the study of the pharmacological qualities of traditional medicinal substances;
- the study of **Phytotherapy** (the medicinal use of plant extracts); and
- **Phytochemistry**, the study of chemicals derived from plants (including the identification of new drug candidates derived from plant sources).
- **Zoopharmacognosy**, the process by which animals self-medicate, by selecting and using plants, soils, and insects to treat and prevent disease.
- **Marine Pharmacognosy**, the study of chemicals derived from marine organisms.

INTRODUCTION

The term comes from two Greek words: "pharmakon" meaning drug or medicine, and "gnosis" meaning knowledge.

Pharmacognosy is "the study of the physical, chemical, biochemical and biological properties of drugs, drug substances or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources".

Pharmacognosy is the study of medicinal uses of various naturally occurring drugs and its history, sources, distributions, method of cultivation, active constituents, medicinal uses, identification test, preservation methods, substituents and adulterants.

Plant preparations are said to be medicinal or herbal when they are used to promote health beyond basic nutrition.

The study of drugs from plants includes the subjects of botany, chemistry and pharmacology.

Botany includes the identification (taxonomy), genetics, and cultivation of plants.

Chemical characterization of includes the isolation, identification and quantification of constituents in plant materials.

Pharmacology is the study of the biological effects that the chemicals in medicinal plants have on cell cultures, animals and humans practical perspectives as follows

- ✓ Quality control (identity, purity, consistency)
- ✓ Efficacy (therapeutic indications, pharmacological investigations)
- ✓ Safety (adverse reactions, drug interactions, contraindications, precautions)

Crude Drugs: -

- Crude drugs are the drugs, which are obtained from natural sources like plant, Animals, minerals, marine, microorganisms and they are used as such as they occur in nature without any processing except, drying & size reduction.
- Crude drugs is referred to the natural product that has not been in advanced in value or improved in condition by any process or treatment beyond that which is essential for its proper packaging and prevention from deterioration.

History of Pharmacognosy:

Drug discovery from natural products have played and continue to play an invaluable role as sources of drugs or lead compounds in the prophylaxis and treatment of diseases.

Plants, especially those with pharmacological uses have been the primary sources of medicines and have an advantage over other sources in drug discovery for various reasons.

'Pharmacognosy'—has been coined by the merger of two Greek words

Pharmakon means Drug and *Gnosis* means knowledge i.e., the knowledge of drugs.

The nomenclature 'Pharmacognosy' was used first and foremost by *C.A. Seydler*,

In about 77 AD, Dioscorides, a Greek doctor, kept a record of about 600 kinds of crude drugs in his compiled book *De Materia Medica*, a book that had played an important role in pharmacology and botany.

In 1815, C.A. Seydler, a German who used the word "pharmakognosie" in his book named *Analecta Pharmacognostica*, was referred to as the father of pharmacognosy.

Work of Galen: - (131 - 200)

Galen was Greek pharmacist; he worked on extraction of chemical constituent from the plants. He developed various methods of extraction therefore the branch of pharmacy which deals with extraction of chemical constituent from plants & animals is called as *Galenical Pharmacy*.

Hippocrates (460 - 360, B.C.)

Before the birth of Jesus. He was Greek scientist; he worked on human anatomy & Physiology Particularly on circulatory system & nervous system. He prepared famous oath for physicians, which is still taken by the physicians. He is known as father of medicine.

Indian History of Pharmacognosy:

It is about 5500 years old, in the suktas of Rigveda & Atharwaveda medicinal property of plant is given. There are several medicinal plants, which are given with their use. The old Ayurveda Books, Charak samhita & Sushrutsamhita described many medicinal plants.

The Scope of Pharmacognosy:

- ✓ Pharmacognosy gives a sound knowledge of the vegetable drugs under botany and animal drugs under zoology.
- ✓ It also includes plant taxonomy, plant breeding, plant pathology, plant genetics and by this knowledge one can improve the cultivation methods for both medicinal and aromatic plants.
- ✓ Nowadays phytochemistry (plant chemistry) has undergone the significant improvement.
- ✓ This includes a variety of substances that are accumulated by plants and synthesized by plants.

A vital contribution to the advancement of natural science:

- ✓ This has done by the advanced technologies of cultivation, purification, identification (characterization) of pharmaceuticals from nature.
- ✓ Concepts of biochemistry and chemical engineering help in the improvement of collection, processing and storage technologies of pharmaceuticals.
- ✓ It also gives knowledge of chemotaxonomy, biogenic pathways for the formation of active ingredients.

A vital link between pharmacology and medicinal chemistry:

- ✓ Newly detected plant drugs are converting into medicine as purified phytochemicals.
- ✓ Pharmacognosy is essential for the evolution of new medicines because crude drugs are used for the preparation of galenicals or as a sources of therapeutically active metabolites.
- ✓ In short Pharmacognosy is an important link between pharmaceuticals and basic science as well as ayurvedic and allopathic system of medicines.
- ✓ Pharmacognosy is a science of active principles of crude drugs and which can be help in dispensing, formulating, and manufacturing of dosage forms.
- ✓ In other way the complete knowledge of Pharmacognosy will help in the recent trend that is in industries,
- ✓ As a research tools and in new drug delivery systems, and all the departments of pharmaceuticals and one can improve the healthcare facilities across the world.